

PROJECT: Data Lake Implementation using AWS Glue and Athena

Services: S3, Glue, Athena, IAM.

Description: Create a data lake on S3, catalog the data using AWS Glue, and perform queries using Athena. Configure IAM policies for secure access.

Skills: Data cataloging, serverless querying, data lake architecture.

Definition:

This project involves creating a centralized data lake on Amazon S3 to store structured and unstructured data. The data is cataloged using AWS Glue, which automatically discovers and organizes the datasets. Amazon Athena is then used to query the data using standard SQL, enabling analytics without the need for managing servers. Access to data and services is secured using IAM roles and policies, ensuring that only authorized users and services can read or write data.

Scope of the Project:

Centralized Storage: Store structured and unstructured data in Amazon S3 for easy management.

Data Cataloging: Automatically discover and organize datasets using AWS Glue Data Catalog.

Serverless Querying: Use Amazon Athena to run SQL queries on data directly in S3 without provisioning servers.

Secure Access: Implement IAM roles and policies to control and secure access to data and services.

Methodologies:

1. **S3 Bucket Setup:** Create S3 buckets to store raw, processed, and curated data.
2. **Data Ingestion:** Upload structured and unstructured data to the S3 data lake.
3. **Glue Crawler:** Configure AWS Glue crawlers to scan S3 data and create the data catalog.
4. **ETL Jobs (Optional):** Use AWS Glue ETL jobs to clean, transform, and organize the data.
5. **Athena Queries:** Run SQL queries on the cataloged data for analytics and reporting.
6. **IAM Security:** Define and attach IAM roles and policies to secure access to S3, Glue, and Athena.
7. **Validation & Testing:** Verify data accessibility, query results, and security permissions.

Services Used in this Project:

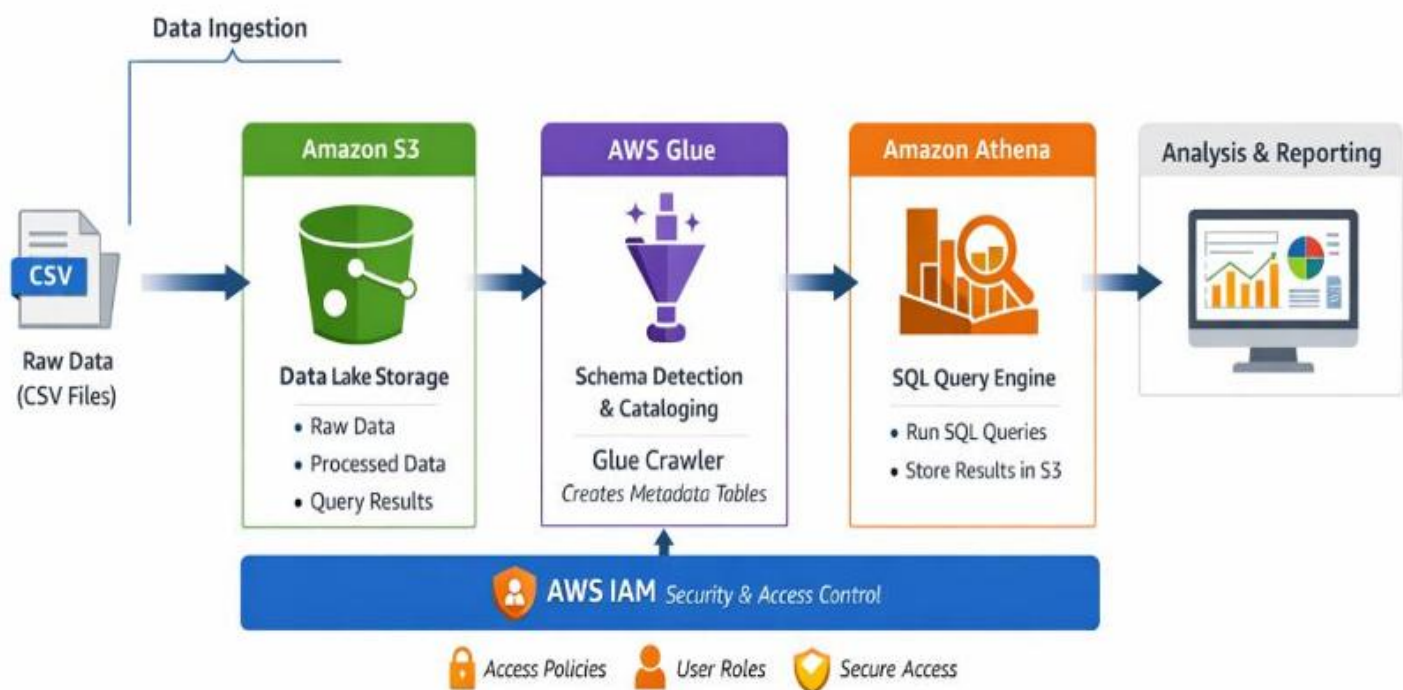
Amazon S3 – For storing raw, processed, and curated data in the data lake.

AWS Glue – For data cataloging, schema discovery, and optional ETL transformations.

Amazon Athena – For serverless SQL queries and analytics on S3 data.

AWS IAM – For managing roles, policies, and secure access to services and data.

Architecture Diagram:

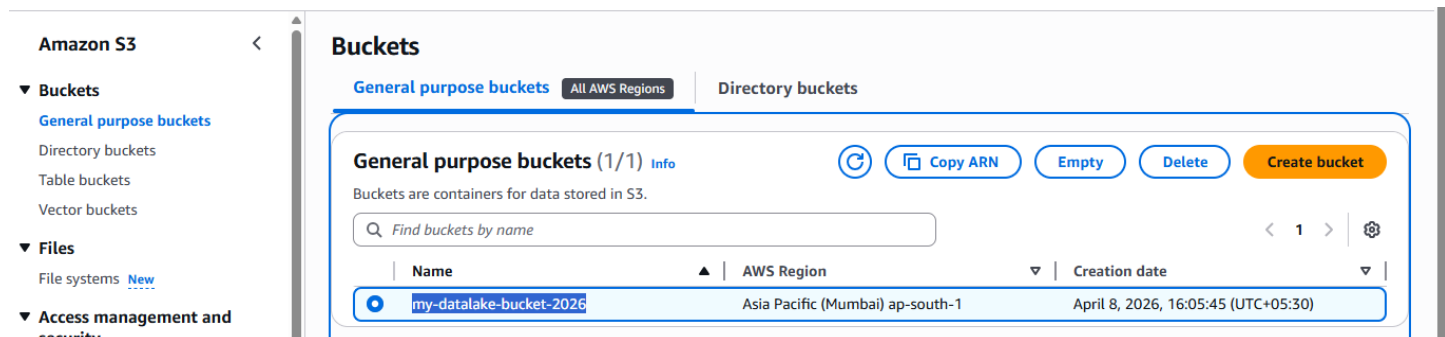


Implementation and Execution of a Data Lake Using AWS Glue and Athena.

Phase 1: Create Data Lake Storage (Amazon S3)

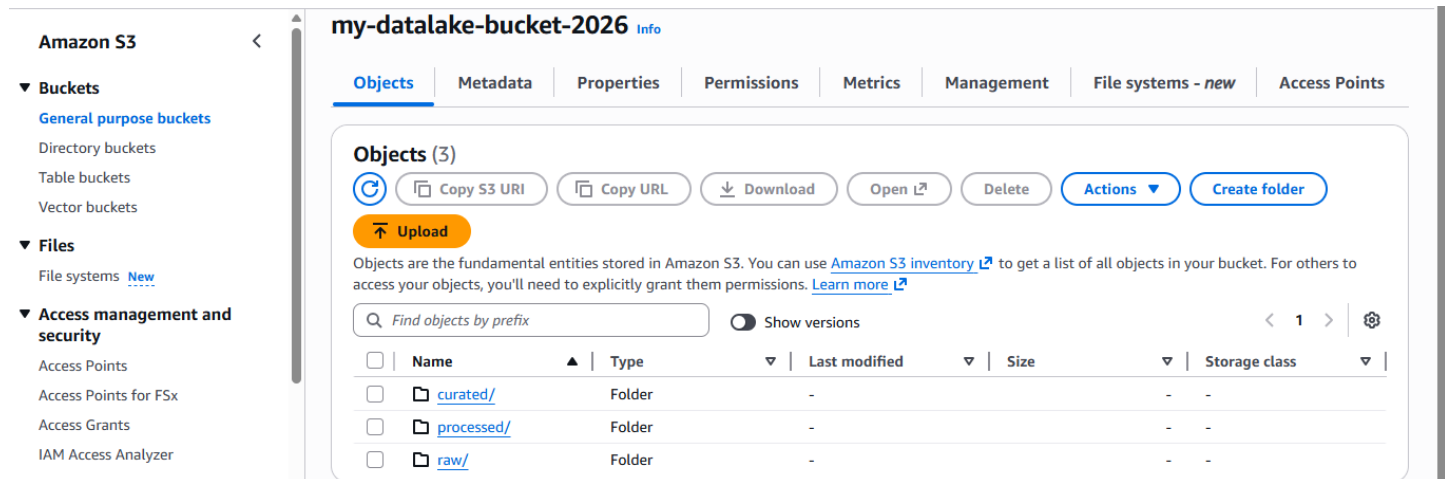
Step 1: Create S3 Bucket

- Bucket Name: my-datalake-bucket-2026
- Region: ap-south-1 (Mumbai)



Step 2: Create Folder Structure Inside Bucket

- Created the following folders: my-datalake-bucket-2026/
 - Raw/
 - Processed/
 - Curated/



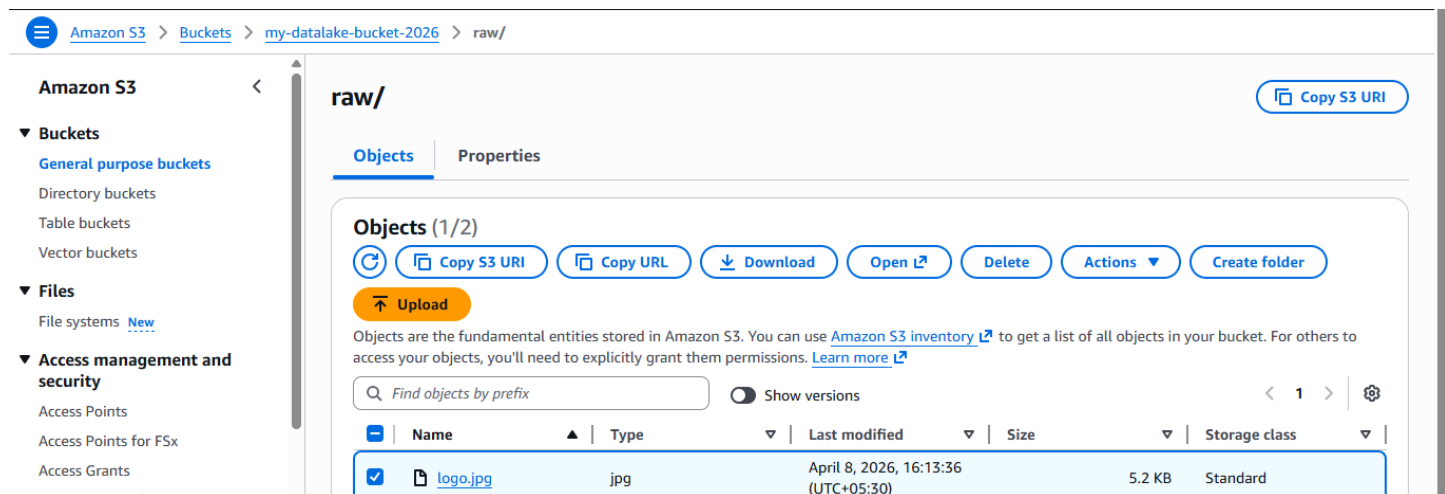
Phase 2: Upload Raw Data

Step 3: Create Sample File

- Created logo.jpg

Step 4: Upload File to S3: Uploaded to

s3://my-datalake-bucket-2026/raw/logo.jpg



Phase 3: Configure Security (IAM)

Step 5: Create IAM Role

Created role: GlueServiceRole

- Attached policies:
 - AmazonS3FullAccess
 - AWSGlueServiceRole

GlueServiceRole info Delete

Allows Glue to call AWS services on your behalf.

Summary Edit

Creation date
April 08, 2026, 16:43 (UTC+05:30)

Last activity
19 hours ago

ARN
arn:aws:iam::074139044318:role/GlueServiceRole

Maximum session duration
1 hour

Permissions **Trust relationships** **Tags** **Last Accessed** **Revoke sessions**

Permissions policies (2) info Simulate Remove Add permissions

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AmazonS3FullAccess	AWS managed	2
AWSGlueServiceRole	AWS managed	2

Phase 4: AWS Glue Setup

Step 6: Create Glue Database

- Database Name: datalake_db

Databases (1/2) Last updated (UTC) April 9, 2026 at 06:45:17 Edit Delete Add database

A database is a set of associated table definitions, organized into a logical group.

Choose catalog: 074139044318

Filter databases

Name	Description	Location URI	Source catalog	Created on (UTC)
☒ datalake_db	-	-	074139044318	April 8, 2026 at 10:52:49

Step 7: Create Glue Crawler

Crawler Name: projectcrawler

- Configured:
 - Data source → s3://my-datalake-bucket-2026/raw/
 - IAM Role → GlueServiceRole
 - Target database → datalake_db
 - Schedule → On demand

projectcrawler Last updated (UTC) April 9, 2026 at 06:43:51 Run crawler Edit Delete

Crawler properties

Name projectcrawler	IAM role GlueServiceRole	Database datalake_db	State READY
Description -	Security configuration -	Lake Formation configuration -	Table prefix -
Maximum table threshold -			

Advanced settings

Data Catalog tables
Data connections
Workflows (orchestration)
▼ Data Catalog
Catalogs
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers

Crawlers

A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.

Crawlers (1/1) Info

Last updated (UTC)
April 9, 2026 at 06:47:27



Action ▼

Run

Create crawler

View and manage all available crawlers.

Filter crawlers

< 1 > ⚙

☒	Name	State	Schedule	Last run	Last run ti...	Log	Table change...
☒	projectcrawler	Ready		✓ Succeeded	April 8, 2026 ...	View log	-

Phase 5: Amazon Athena Setup

Step 9: Configure Query Result Location

- Set in Athena: `s3://my-datalake-bucket-2026/query-results/`



Query in Amazon SageMaker Unified Studio

Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment.

Set up [↗](#)



Editor

Recent queries

Saved queries

Settings

Workgroup

primary ▼

Query result encryption

Manage

Query result location

[s3://my-datalake-bucket-2026/raw/](#) [↗](#)

Encrypt query results

-

Expected bucket owner

-

Assign bucket owner full control over query results
Turned off

Step 10: Run SQL Query

Output displayed in Athena:

Athena now supports typeahead code suggestions to speed up SQL query development

Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.

Edit preferences



Data

Data source

AwsDataCatalog

Catalog

None

Database

datalake_db

Tables and views

Create

Filter tables and views

Tables (2)

shree_csv

unsaved

Partitioned

Views (0)

Query 1 : X Query 2 : X Query 3 : X Query 4 : X

```
1 SELECT *  
2 FROM shree_csv  
3 LIMIT 10;
```

SQL Ln 3, Col 10

Run again

Explain [↗](#)

Cancel

Clear

Create

Reuse query results
up to 60 minutes ago

Query results

Query stats

Completed

Time in queue: 57 ms

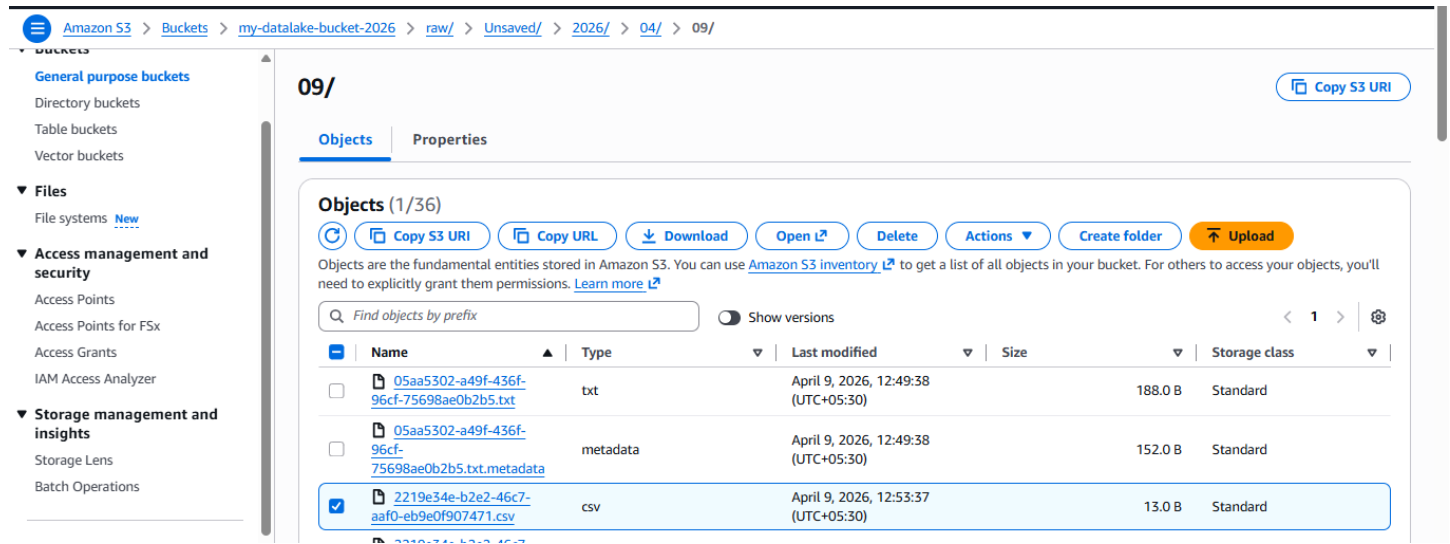
Run time: 323 ms

Data scanned: -

Phase 6: Final Output Storage:

- Athena stored query output in S3:

s3://my-datalake-bucket-2026/raw/Unsaved/2026/04/09/unique ID.csv



The screenshot shows the Amazon S3 console interface. The breadcrumb navigation at the top indicates the path: Amazon S3 > Buckets > my-datalake-bucket-2026 > raw/ > Unsaved/ > 2026/ > 04/ > 09/. On the left sidebar, the 'Buckets' section is expanded, showing 'General purpose buckets' with sub-items 'Directory buckets', 'Table buckets', and 'Vector buckets'. Below this, the 'Files' section is expanded, showing 'File systems' with a 'New' button. The 'Access management and security' section is also expanded, showing 'Access Points', 'Access Points for FSx', 'Access Grants', and 'IAM Access Analyzer'. The 'Storage management and insights' section is expanded, showing 'Storage Lens' and 'Batch Operations'. The main content area shows the '09/' directory. The 'Objects' tab is selected, displaying a list of three objects. The first two objects are .txt files, and the third is a .csv file, which is selected. The table columns are: Name, Type, Last modified, Size, and Storage class. The selected .csv file has a size of 13.0 B and is stored in the Standard storage class.

Name	Type	Last modified	Size	Storage class
05aa5302-a49f-436f-96cf-75698ae0b2b5.txt	txt	April 9, 2026, 12:49:38 (UTC+05:30)	188.0 B	Standard
05aa5302-a49f-436f-96cf-75698ae0b2b5.txt.metadata	metadata	April 9, 2026, 12:49:38 (UTC+05:30)	152.0 B	Standard
2219e34e-b2e2-46c7-aaf0-eb9e0f907471.csv	csv	April 9, 2026, 12:53:37 (UTC+05:30)	13.0 B	Standard

Troubleshoot:

- Glue Table Name Created as "raw"

Later I changed to "shree"

Conclusion:

This project successfully demonstrates how to build a serverless and scalable data lake on AWS using Amazon S3, AWS Glue, and Amazon Athena. We stored raw data in S3, automatically detected its schema using Glue, and analyzed it using SQL queries in Athena without managing any servers. The project highlights secure access control using IAM and shows how query results can be stored for reporting and further analysis. Overall, it proves that AWS provides a simple, cost-effective, and efficient way to manage and analyze data in the cloud.